

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 May/June 2022 Worked Solutions

May/June 2022 paper rebuilt with the worked solution after each question.

10

paper questions

WMA11

Pure 1

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P1**

PAST PAPER

WMA11/01 May/June 2022

May/June 2022 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Integration

1. Find

$$\int \left(10x^5 + 6x^3 - \frac{3}{x^2} \right) dx$$

giving your answer in simplest form.

(4)

(Total 4 marks)

Worked Solution - Question 1

1. Rewrite the negative power

$$\int \left(10x^5 + 6x^3 - \frac{3}{x^2} \right) dx = \int (10x^5 + 6x^3 - 3x^{-2}) dx.$$

2. Integrate term by term

$$\int 10x^5 dx = \frac{10x^6}{6} = \frac{5}{3}x^6, \int 6x^3 dx = \frac{6x^4}{4} = \frac{3}{2}x^4, \text{ and } \int -3x^{-2} dx = 3x^{-1}.$$

3. Add the constant

Therefore the integral is $\frac{5}{3}x^6 + \frac{3}{2}x^4 + \frac{3}{x} + c$.

Final answer

$$\frac{5}{3}x^6 + \frac{3}{2}x^4 + \frac{3}{x} + c.$$

Question 2

Basic Trigonometry

2. In the triangle ABC ,

- $AB = 21$ cm
- $BC = 13$ cm
- angle $BAC = 25^\circ$
- angle $ACB = x^\circ$

(a) Use the sine rule to find the value of $\sin x^\circ$, giving your answer to 4 decimal places. (2)

Given also that AB is the longest side of the triangle,

(b) find the value of x , giving your answer to 2 decimal places. (3)

(Total 5 marks)

Worked Solution - Question 2

1. Use the sine rule

The side opposite x is $AB = 21$, and the side opposite 25° is $BC = 13$, so

$$\frac{\sin x}{21} = \frac{\sin 25^\circ}{13}.$$

2. Find the sine value

$$\sin x = \frac{21 \sin 25^\circ}{13} = 0.68269 \dots$$

3. Choose the correct angle

The angle from the calculator is 43.05° , but AB is the longest side, so x is obtuse.

Hence $x = 180^\circ - 43.05^\circ = 136.95^\circ$.

Final answer

$x = 137^\circ$ to 3 significant figures.

Question 3

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

- (i) Show that $\frac{\sqrt{180} - \sqrt{80}}{\sqrt{5}}$ is an integer and find its value.

(2)

- (ii) Simplify

$$\frac{4\sqrt{5} - 5}{7 - 3\sqrt{5}}$$

giving your answer in the form $a + b\sqrt{5}$ where a and b are rational numbers.

(3)

(Total 5 marks)

Worked Solution - Question 3

1. Simplify the square roots

$$\sqrt{180} = 6\sqrt{5} \text{ and } \sqrt{80} = 4\sqrt{5}.$$

2. Complete part (a)

$$\frac{\sqrt{180} - \sqrt{80}}{\sqrt{5}} = \frac{6\sqrt{5} - 4\sqrt{5}}{\sqrt{5}} = 2.$$

3. Rationalise part (b)

$$\frac{4\sqrt{5} - 5}{7 - 3\sqrt{5}} \times \frac{7 + 3\sqrt{5}}{7 + 3\sqrt{5}} = \frac{(4\sqrt{5} - 5)(7 + 3\sqrt{5})}{49 - 45}.$$

4. Expand and simplify

The numerator is $28\sqrt{5} + 60 - 35 - 15\sqrt{5} = 25 + 13\sqrt{5}$, so the expression is $\frac{25 + 13\sqrt{5}}{4}$.

Final answer

(a) 2.

(b) $\frac{25}{4} + \frac{13}{4}\sqrt{5}$.

Question 4

Transformations

4.

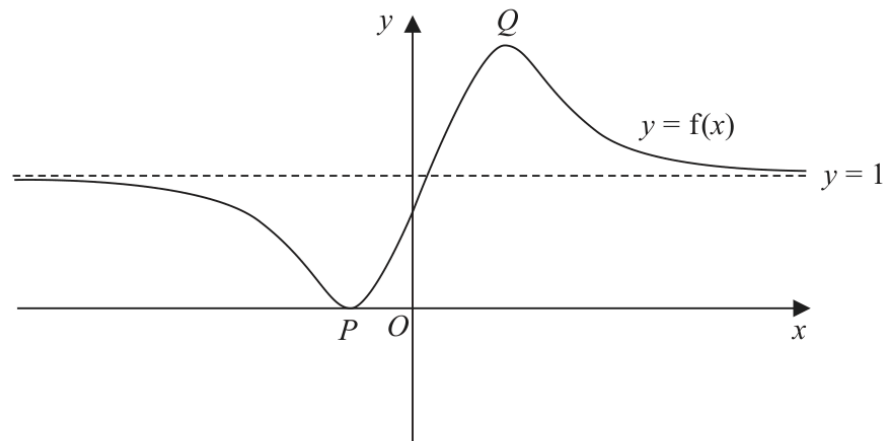


Figure 1

Figure 1 shows a sketch of a curve with equation $y = f(x)$

The curve has a minimum at $P(-1, 0)$ and a maximum at $Q\left(\frac{3}{2}, 2\right)$

The line with equation $y = 1$ is the only asymptote to the curve.

On separate diagrams sketch the curves with equation

(i) $y = f(x) - 2$ (3)

(ii) $y = f(-x)$ (3)

On each sketch you must clearly state

- the coordinates of the maximum and minimum points
- the equation of the asymptote

(Total 6 marks)

Worked Solution - Question 4

1. Translate for $y = f(x) - 2$

Subtracting 2 moves every point of the graph down by 2 units.

2. State the new key features

The minimum $P(-1, 0)$ becomes $(-1, -2)$, the maximum $Q(\frac{3}{2}, 2)$ becomes $(\frac{3}{2}, 0)$, and the asymptote $y = 1$ becomes $y = -1$.

3. Reflect for $y = f(-x)$

Replacing x by $-x$ reflects the graph in the y-axis.

4. State the reflected key features

The minimum $P(-1, 0)$ becomes $(1, 0)$, the maximum $Q(\frac{3}{2}, 2)$ becomes $(-\frac{3}{2}, 2)$, and the horizontal asymptote remains $y = 1$.

Final answer

(i) minimum $(-1, -2)$, maximum $(\frac{3}{2}, 0)$, asymptote $y = -1$. (ii) minimum $(1, 0)$, maximum $(-\frac{3}{2}, 2)$, asymptote $y = 1$.

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9 marks

Question 5

Straight Line

5. The curve C has equation $y = f(x)$

Given that

- $f(x)$ is a quadratic expression
- the maximum turning point on C has coordinates $(-2, 12)$
- C cuts the negative x -axis at -5

- (a) find $f(x)$

(4)

The line l_1 has equation $y = \frac{4}{5}x$

Given that the line l_2 is perpendicular to l_1 and passes through $(-5, 0)$

- (b) find an equation for l_2 , writing your answer in the form $y = mx + c$ where m and c are constants to be found.

(3)

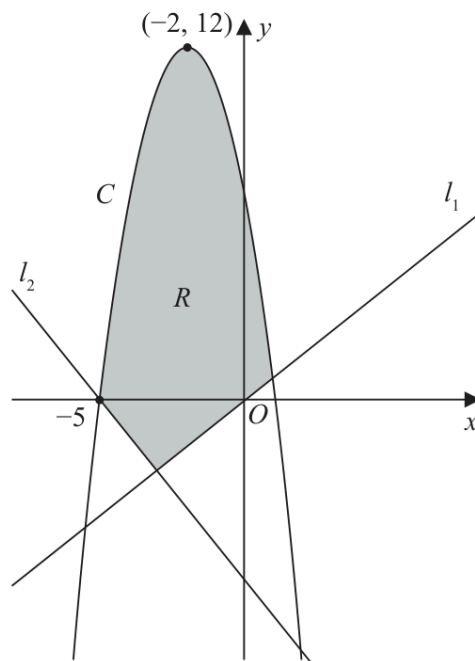


Figure 2

Figure 2 shows a sketch of the curve C and the lines l_1 and l_2

- (c) Define the region R , shown shaded in Figure 2, using inequalities.

(2)

(Total 9 marks)

Worked Solution - Question 5

1. Use symmetry of the quadratic

The maximum is at $x = -2$, and one root is $x = -5$. The other root is the same distance to the right of -2 , so it is $x = 1$.

2. Find the quadratic

Let $f(x) = a(x + 5)(x - 1)$. Since the maximum point is $(-2, 12)$,
 $12 = a(3)(-3) = -9a$, so $a = -\frac{4}{3}$.

3. Find the perpendicular gradient

The gradient of l_1 is $\frac{4}{5}$, so the gradient of a perpendicular line is $-\frac{5}{4}$.

4. Find l_2

Using the point $(-5, 0)$, $y = -\frac{5}{4}x + c$. Then $0 = \frac{25}{4} + c$, so $c = -\frac{25}{4}$ and l_2 is
 $y = -\frac{5}{4}x - \frac{25}{4}$.

5. Write the inequalities for R

The region is below the curve and above both straight lines, so $y \leq f(x)$,
 $y \geq \frac{4}{5}x$, and $y \geq -\frac{5}{4}x - \frac{25}{4}$.

Final answer

$$(a) f(x) = -\frac{4}{3}(x + 5)(x - 1).$$

$$(b) y = -\frac{5}{4}x - \frac{25}{4}.$$

$$(c) y \leq f(x), y \geq \frac{4}{5}x, y \geq -\frac{5}{4}x - \frac{25}{4}.$$

Question 6

Quadratics

6. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

- (a) Given that

$$2xy - 3x^2 = 50$$

and

$$y - x^3 + 6x = 0$$

show that

$$2x^4 - 15x^2 - 50 = 0 \quad (2)$$

- (b) Hence solve the simultaneous equations

$$2xy - 3x^2 = 50$$

$$y - x^3 + 6x = 0$$

Give your answers in fully simplified surd form.

(5)

(Total 7 marks)

Worked Solution - Question 6

1. Rearrange the second equation

From $y - x^3 + 6x = 0$, we get $y = x^3 - 6x$.

2. Substitute into the first equation

$2x(x^3 - 6x) - 3x^2 = 50$, so $2x^4 - 15x^2 - 50 = 0$.

3. Solve the quadratic in x squared

Let $u = x^2$. Then $2u^2 - 15u - 50 = 0$, giving $u = 10$ or $u = -\frac{5}{2}$.

4. Reject the negative value

Since x^2 cannot be negative, $x^2 = 10$, so $x = \pm\sqrt{10}$.

5. Find y

$y = x^3 - 6x = x(x^2 - 6) = x(10 - 6) = 4x$. Therefore $y = 4\sqrt{10}$ when $x = \sqrt{10}$ and $y = -4\sqrt{10}$ when $x = -\sqrt{10}$.

Final answer

$(x, y) = (\sqrt{10}, 4\sqrt{10})$ or $(-\sqrt{10}, -4\sqrt{10})$.

Question 7

Integration

7. The curve C has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = \frac{2}{\sqrt{x}} + \frac{A}{x^2} + 3$, where A is a constant
- $f''(x) = 0$ when $x = 4$

(a) find the value of A .

(4)

Given also that

- $f(x) = 8\sqrt{3}$, when $x = 12$

(b) find $f(x)$, giving each term in simplest form.

(5)

(Total 9 marks)

Worked Solution - Question 7

1. Differentiate $f'(x)$

$$f'(x) = 2x^{-1/2} + Ax^{-2} + 3, \text{ so } f''(x) = -x^{-3/2} - 2Ax^{-3}.$$

2. Use $f''(4) = 0$

$$-4^{-3/2} - 2A(4^{-3}) = 0, \text{ so } -\frac{1}{8} - \frac{A}{32} = 0 \text{ and } A = -4.$$

3. Substitute A into $f'(x)$

$$f'(x) = \frac{2}{\sqrt{x}} - \frac{4}{x^2} + 3.$$

4. Integrate to find $f(x)$

$$f(x) = 4\sqrt{x} + \frac{4}{x} + 3x + c.$$

5. Use $f(12) = 8\sqrt{3}$

$$f(12) = 8\sqrt{3} + \frac{1}{3} + 36 + c = 8\sqrt{3}, \text{ so } c = -\frac{109}{3}.$$

Final answer

$$f(x) = 4\sqrt{x} + \frac{4}{x} + 3x - \frac{109}{3}.$$

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10 marks

Question 8

Radians

8.

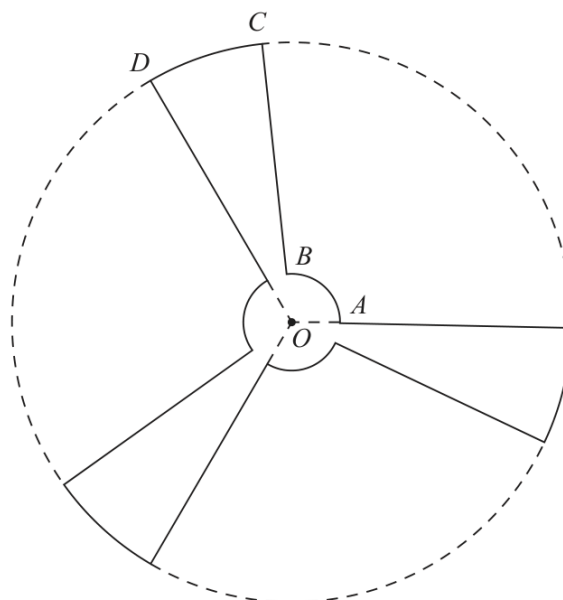


Figure 3

Figure 3 shows a sketch of the outline of the face of a ceiling fan viewed from below.

The fan consists of three identical sections congruent to $OABCD$, shown in Figure 3, where

- $OABO$ is a sector of a circle with centre O and radius 9 cm
- $OBCDO$ is a sector of a circle with centre O and radius 84 cm
- angle $AOD = \frac{2\pi}{3}$ radians

Given that the length of the arc AB is 15 cm,

- (a) show that the length of the arc CD is 35.9 cm to one decimal place. (3)

The face of the fan is modelled to be a flat surface.

Find, according to the model,

- (b) the perimeter of the face of the fan, giving your answer to the nearest cm, (2)
- (c) the surface area of the face of the fan.

Give your answer to 3 significant figures and make your units clear. (5)

(Total 10 marks)

Worked Solution - Question 8

1. Find the small sector angle

For arc AB , $s = r\theta$, so $15 = 9\theta$ and $\theta = \frac{5}{3}$ radians.

2. Find the outer arc angle

Each identical section has total angle $\frac{2\pi}{3}$, so the angle for arc CD is

$$\frac{2\pi}{3} - \frac{5}{3} = \frac{2\pi - 5}{3}.$$
3. Find CD

$$CD = 84 \left(\frac{2\pi - 5}{3} \right) = 28(2\pi - 5) = 35.9 \text{ cm to 3 significant figures.}$$

4. Find the perimeter

There are 3 inner arcs, 3 outer arcs, and 6 straight radial edges of length $84 - 9 = 75$. Thus $P = 3(15) + 3(35.929\dots) + 6(75) = 602.8\dots \text{ cm}$.

5. Find the area of one section

One section has area $\frac{1}{2}(9^2) \left(\frac{5}{3} \right) + \frac{1}{2}(84^2) \left(\frac{2\pi - 5}{3} \right) = 1576.525\dots \text{ cm}^2$.

6. Find the total area

The total area is $3(1576.525\dots) = 4729.57\dots \text{ cm}^2 = 4.73 \times 10^3 \text{ cm}^2$ to 3 significant figures.

Final answer

(a) $CD = 35.9 \text{ cm.}$

(b) 603 cm.

(c) $4.73 \times 10^3 \text{ cm}^2.$

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8 marks

Question 9

Trigonometric Functions

9.

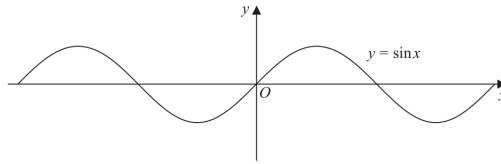


Figure 4

Figure 4 shows part of the graph of the curve with equation $y = \sin x$

Given that $\sin \alpha = p$, where $0 < \alpha < 90^\circ$

- (a) state, in terms of p , the value of

- (i) $2 \sin(180^\circ - \alpha)$
(ii) $\sin(\alpha - 180^\circ)$
(iii) $3 + \sin(180^\circ + \alpha)$

(3)

A copy of Figure 4, labelled Diagram 1, is shown on page 27.

On Diagram 1,

- (b) sketch the graph of $y = \sin 2x$

(2)

- (c) Hence find, in terms of α , the x coordinates of any points in the interval $0 < x < 180^\circ$ where

$$\sin 2x = p$$

(3)

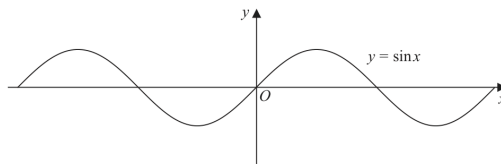


Diagram 1

(Total 8 marks)

Worked Solution - Question 9

1. Use sine symmetry

$$\sin(180^\circ - \alpha) = \sin \alpha = p, \text{ so } 2 \sin(180^\circ - \alpha) = 2p.$$

2. Use odd symmetry

$$\sin(\alpha - 180^\circ) = -\sin(180^\circ - \alpha) = -p.$$

3. Use the third quadrant identity

$$\sin(180^\circ + \alpha) = -\sin \alpha = -p, \text{ so } 3 + \sin(180^\circ + \alpha) = 3 - p.$$

4. Sketch $y = \sin 2x$

The graph has amplitude 1 and period 180° . It crosses the x-axis at $0^\circ, 90^\circ, 180^\circ$, has a maximum at 45° , and a minimum at 135° .

5. Solve $\sin 2x = p$

Since $\sin \alpha = p$ and $0 < 2x < 360^\circ$, the possible values are $2x = \alpha$ or $2x = 180^\circ - \alpha$.

6. Find x

$$\text{Therefore } x = \frac{\alpha}{2} \text{ or } x = 90^\circ - \frac{\alpha}{2}.$$

Final answer

$$(a)(i) 2p, (ii) -p, (iii) 3 - p.$$

$$(c) x = \frac{\alpha}{2}, 90^\circ - \frac{\alpha}{2}.$$

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12 marks

Question 10

Differentiation

10.

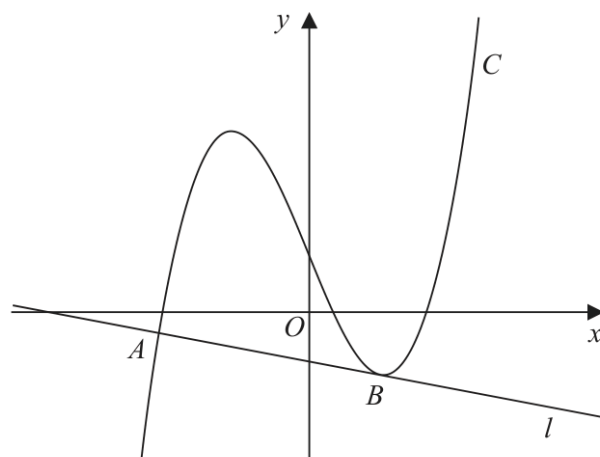


Figure 5

Figure 5 shows a sketch of the curve C with equation

$$y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$$

where k is a constant.

- (a) Find $\frac{dy}{dx}$ (2)

The line l , shown in Figure 5, is the normal to C at the point A with x coordinate $-\frac{7}{2}$

Given that l is also a tangent to C at the point B ,

- (b) show that the x coordinate of the point B is a solution of the equation

$$12x^2 + 4x - 33 = 0$$

(4)

- (c) Hence find the x coordinate of B , justifying your answer. (2)

Given that the y intercept of l is -1

- (d) find the value of k . (4)

(Total 12 marks)

Worked Solution - Question 10

1. Differentiate the curve

For $y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$, $\frac{dy}{dx} = \frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2}$.

2. Find the tangent gradient at A

At $x = -\frac{7}{2}$, $\frac{dy}{dx} = \frac{6}{7}\left(\frac{49}{4}\right) + \frac{2}{7}\left(-\frac{7}{2}\right) - \frac{5}{2} = \frac{21}{2} - 1 - \frac{5}{2} = 7$.

3. Find the gradient of l

Since l is the normal at A , its gradient is the negative reciprocal of 7, so

$$m_l = -\frac{1}{7}.$$

4. Use that l is tangent at B

At B , the tangent gradient is also $-\frac{1}{7}$, so $\frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2} = -\frac{1}{7}$.

5. Reach the given quadratic

Multiplying by 14 gives $12x^2 + 4x - 35 = -2$, hence $12x^2 + 4x - 33 = 0$.

6. Solve for the x-coordinate of B

$12x^2 + 4x - 33 = 0$ gives $x = \frac{-4 \pm 40}{24}$, so $x = \frac{3}{2}$ or $x = -\frac{11}{6}$. From the diagram, B is the point to the right of the y-axis, so $x_B = \frac{3}{2}$.

7. Use the y-intercept of l

Since the y-intercept is -1 and the gradient is $-\frac{1}{7}$, the line is $y = -\frac{1}{7}x - 1$.

8. Find the y-coordinate of A

At $x = -\frac{7}{2}$ on the line, $y = -\frac{1}{7}\left(-\frac{7}{2}\right) - 1 = \frac{1}{2} - 1 = -\frac{1}{2}$.

9. Substitute A into the curve

$$-\frac{1}{2} = \frac{2}{7} \left(-\frac{7}{2}\right)^3 + \frac{1}{7} \left(-\frac{7}{2}\right)^2 - \frac{5}{2} \left(-\frac{7}{2}\right) + k = -\frac{7}{4} + k.$$

10. Find k

Therefore $k = \frac{5}{4}$.

Final answer

(a) $\frac{dy}{dx} = \frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2}.$

(c) $x_B = \frac{3}{2}.$

(d) $k = \frac{5}{4}.$